

Encore Computer Series 91 & 93
Proposed 1991-1993 Market Strategy For
Financial and Information Services Industries

Objective and Methodology:

Aberdeen Group has been asked to advise Encore Computer Corporation on potential merchandising strategies and themes to increase Encore's marketshare for the justshipped Series 91 and the soon-to-be-shipped Series 93 within financial and information services organizations. For purposes of this report, financial organizations include banks and brokerage houses and information services include enterprises that sell online information.

Aberdeen Group conducts on average 25 user interviews each month. Based upon the history of these interviews, the experience and expertise of Aberdeen's principals, and extensive discussions with Encore executives, Aberdeen recommends the following merchandising strategy.

Our objectives in this report include describing for the Series 91 & 93 in both the financial and information services marketplaces:

- market opportunity, size and growth
- recommended market strategy
- key positioning and differentiating messages
- product issues: base, specials & futures
- distribution program recommendations
- competitive overview
- sales program recommendations

We believe that Encore sales, public relations and marketing communications groups should be able to use this document as a basis for gaining awareness of both Encore's and the Series 91 & 93's capabilities to meet the needs of financial and information services enterprises. However, we wish to note that this document may be relatively difficult to comprehend on a first reading. Not only is the document the embodiment of multiple meetings between Encore and Aberdeen but much of the basis for recommendations that are stated in the beginning are discussed in more depth at the end. Unlike a good mystery novel, we would prefer to begin with the "who did it" and leave the why and how until the ending. And like a good executive report, we have emphasized conclusions and recommendations over long, drawn out proof theorems.

Aberdeen encourages any Encore executives who have questions about this report after reading it to contact Aberdeen Group principals Peter Kastner or John Logan directly.

Finally, the overall objective of this merchandising plan is to aid Encore's field sales and support organization by creating awareness of the unique capabilities of the Series 91 & 93 among MIS decision makers at leading-edge financial and information services organization as well as the independent systems integrators that work within both industries.

Executive Summary

Introduction

Encore has the opportunity with the Series 91 and 93 to establish a new architectural standard in the financial and information services markets during the 1990s. While Encore has the option to profitlessly compete in the commodity Risc/Open Systems product space like so many other suppliers, the financial and information services will evaluate an Encore "Dirty Computing"-like alternative approach to data center services which yields high availability, superior expansion and investment protection

coupled with multiple-better price-performance than traditional midrange and mainframe alternatives.

A Vision of "Dirty Computing"

The flexibility, expandability, and availability of what Encore has internally called "Dirty Computing", coupled with outstanding potential price/performance, leads Aberdeen to conclude the key buyer catch-phrase could be Adaptive Computing. While Reflective Memory is the key uniqueness we believe Encore should emphasize in any technology discussions, Encore's Adaptive Computing benefits chiefly from the synergy of its constituent parts: symmetric multiprocessing; clustering; memorycoupling; single-system view; high availability.

Adaptive Computing has the features and benefits of an architecture which has the potential staying power to last through the seven-year life-cycle buyer's envision — effectively the rest of the 1990s. Importantly, bundling multiple packaged Series 9X systems connected under Reflective Memory adds substantial value which commands a higher price.

Market Overview

Aberdeen Group sizes the worldwide market for midrange and mainframe systems hardware and software in the financial and information services markets at \$11.3 billion in 1990. We project a 19% annual growth rate — considerably above the industry growth rate — which propels these two markets to about \$20 billion in 1994.

Our research indicates the key buying reasons for the financial market are:

- Reliability;

- Availability;
- Expandability;
- Performance;
- Investment protection;
- Applications; ● Price.

Poor financial performance is driving the financial market to invest more heavily in automation as a capital replacement for labor, and to gain a competitive advantage in the extremely time-critical application of securities trading and some banking applications. Given the high pricing of IBM mainframes, our research shows continued downsizing as new and rearchitected applications appear on lower-cost platforms.

Encore's strengths in the financial industry will be the innovative technology of Adaptive Computing, coupled with attractive price/performance. Encore's weaknesses to market entry include lack of an installed, referencable financial market base, a presently weak end-user support program, and the lack of a sales force fluent in the business of the financial industry.

The information services market is highly fragmented and will be difficult to attack with a comprehensive plan. The fragmentation means there are numerous information service providers whose services are highly specialized and focused on service delivery to their own chosen market. In short, there are no direct marketing vehicles such as trade shows or publications which Encore can use to launch a broad-based marketing program.

Our informal research on Adaptive Computing in the information services market suggests the large market players such as ADP, CompuServe, and Dow Jones will want to evaluate Encore's technology. We believe Encore's strengths in the information services market will relate to the architecture, particularly high availability

and expandability through clustering. Pricing is a higher-weight factor than in the financial markets. Encore's weaknesses include lack of industry presence and an unproven high-availability story.

Both the financial and information services share a number of common threads we believe should be exploited by Encore:

- Applications are time-critical — availability is paramount;
- Applications are communications- and database-intensive;
- These markets are too broad to attack frontally. Use niches;
- Mainframe sites can be readily identified and targeted;
- Applications are being rearchitected to gain a competitive advantage. A new architecture launched under the title Adaptive Computing may be timely.

Market Positioning Recommendations

Most open systems suppliers are claiming the best price/performance in open systems to a largely luke-warm marketplace. Few are making money with this approach and we are not enthusiastic about long-term profitability for any supplier using this approach.

We recommend positioning the Series 91 and 93 with Adaptive Computing as next generation products which can provide the advantages of midrange to mainframe loosely-coupled systems through Encore's unique Reflective Memory. Adaptive Computing can be positioned as a superior computing alternative to the IBM System/390 with DB2, Teradata, expected clustered products from Sequent, Digital's high-end VAXclusters and Tandem. With this positioning, Encore can use its historic strength to sell superior technology to technology buyers.

This positioning posits Encore as the superior alternative to high-end midrange and traditional mainframe platforms, and we believe Encore can justify a steep price premium compared to the generic Risc/open systems market in exchange for a high level of customer support services.

Aberdeen suggests that Encore merchandise the Series 91 and 93 to the financial and information services markets by emphasizing the superior technology of Adaptive Computing for information retrieval and transaction processing.

Coordination of Merchandising and Field Sales

Encore management should consider the following issues as they relate to field selling into the financial and information services markets:

- The Encore sales force has little experience in these markets;
- Encore must be prepared to be the prime contractor in systems integration contracts for the financial and information services markets, as well as providing an umbrella of support services.

Proposed Series 91 and 93 Messages

Aberdeen Group recommends Encore introduce Adaptive Computing as the style of computing that allows MIS managers to quickly and efficiently modify their datacenter systems to rapidly accommodate the changing needs of their enterprises.

Differentiation is required, and Encore should emphasize the technical capabilities and user-enterprise advantages of building datacenter systems in clusters coordinated through RMS. RMS provides a superior vehicle for cross-cluster application integration, and should be contrasted with relatively weak open systems cluster offerings expected by Sequent and others.

Product Issues

Base-level Series 91 and 93 products are hot boxes for commercial markets which rarely buy hot boxes. Packaging SMP CPU nodes and 91-driven I/O channel nodes connected by RMS can be value priced to the commercial markets at attractive margins while at the same time avoiding cannibalization of Encore's traditional, component-based configuration and pricing.

We recommend Encore establish appropriate small, medium, and large versions of Adaptive Computing platforms so the sales force has a packaged configuration to discuss, rather than just the erector set of Series 91 components. We suggest a fully configured two-node low-end system is appropriate to the financial and information services markets at a \$250K price point, and that a high-end system in the \$5 million range is also appropriate, with a clear expansion growth path in between.

Encore will face a tremendous demand in the financial and information services markets for communications products. The company should be prepared to both invest in product development and to work with third-party suppliers.

The biggest application of Encore systems in the commercial sector could be in database-intensive applications, primarily in conjunction with a relational database (RDBMS) such as ASK Ingres or Oracle. To do this successfully, the company must be prepared to optimize each RDBMS for operations on an RMS cluster. Otherwise, the potential advantages of specialized computing- and I/O-processing nodes will be lost. The proof of these efforts should be demonstrated with industry-standard benchmarks such as TPC-A and TPC-B. While the investment will be high, we recommend the company set a goal of 1,000 or 2,000 transactions per second as a demonstration of the robustness of Adaptive Computing.

Distribution Tactics

We see no alternative to direct sales distribution for the Encore Series 90 in the financial and information services markets. We recommend the company hire industry-experienced selling managers to boot-strap sales into the financial markets.

Early selling should focus on winning targeted accounts who will be deliberately oversupported in order to turn these accounts into the industry references Encore will need by 1993.

Competitive Overview

In the financial industry, the IBM System/390 running DB2 is the leader-to-beat. Our research shows buyers are displeased with DB2 performance, resource consumption, and the ability to support more than 100 users. While IBM's grip on the financial markets is strong, they are vulnerable in the RDBMS area where Encore can provide an alternative with multiples-better price-performance.

Tandem is the key target in the information services market. With a cost structure that's propped up by IBM's price levels, Tandem is vulnerable to open systems alternatives in general. However, the company offers superior loosely-coupled cluster technology together with fault tolerance. Encore has the unique opportunity to use the superior clustering of Adaptive Computing against Tandem's clusters, while simultaneously avoiding head-to-head competition on the fault-tolerance issue. Digital's 20,000 VAXclusters are primarily composed of low-performance VAXes. The well-known problems of Digital's distributed lock manager can be contrasted with the engineering elegance of reflective memory for cross-cluster communications.

Teradata has been successful in the financial industry by selling million-dollar RDBMS backends for decision support applications. Their technology is proprietary and based on low-power Intel 80X86 processors. Encore has the opportunity to demonstrate a superior and better-configurable RDBMS backend just as Teradata has succeeded in

generating market demand for an RDBMS backend alternative to traditional midrange and mainframe approaches.

Sequent has achieved modest success in the financial and information services markets, but Sequent sells primarily at the \$100K - \$150K price band. Further, the company lacks a robust cluster strategy. Thus, Encore can go toe-to-toe with Sequent in SMP and win hands down with RMS-based clusters.

Sales Program Recommendations

Initial awareness efforts should be focused in general technical and specific financial technical journals such as the Wall Street Computing Review.

The most effective selling vehicle will be, we believe, a prepackaged half-day seminar geared to selling across individual organizations. A follow-on fifteen minute videotape will help address initial buying objections.

A Vision for "Dirty Computing"

Aberdeen has thoughtfully considered one of the key hurdles Encore will face:
What is it Encore is selling and how is it different or unique?

We recommend that Encore consider the name Adaptive Computing as a way to uniquely begin to describe the compelling attributes of Encore's offering. We use the term Adaptive Computing throughout this report to mean what has been called Dirty Computing internally at Encore.

What is Adaptive Computing?

As detailed in the section on Market Positioning, Aberdeen recommends that Encore focus on the technology of Adaptive Computing as the company's initial plan. This leads to the question, "What is Adaptive Computing?".

Our vision of Adaptive Computing is a technology with a compelling architecture built on a sound, open-systems framework. We believe the key technology elements which must be clearly articulated by Encore marketing and sales are:

- Clustering
- Memory-coupling
- Symmetric Multiprocessing
- Single-system view
- High availability

Encore's Dirty Computing is the only architecture with the combined and simultaneous benefits of symmetric multiprocessing, clustering, memory-coupling, single-system view, and high availability. It is the synergy of these attributes which makes the Encore offering highly attractive for large-scale commercial processing.

To buyers, the adaptiveness of Encore's means it can be applied:

- Flexibly - as the application dictates;
- Redundantly - to supply the high availability crucial to the financial and information services markets;
- Reactively - with easy compute or I/O expansion to accommodate an almost infinite growth path — a powerful benefit to markets where entire new applications are being devised daily;
- Economically - with the power of mainframes but using the price-attractiveness of open systems.

Symmetric Multiprocessing (SMP)

An SMP architecture allows one-to-many processors to operate in parallel under the control of a single OS image. By adding processors, SMP can easily expand vertically. The OS efficiently dispatches work to as many processors as are available. SMP is considered an effective technique for minimizing response times and maximizing throughput by minimizing the time a task spends waiting for CPU cycles.

Clustering

While SMP systems can show excellent performance growth, horizontal growth also has benefits. Clustered systems are composed of multiple processor nodes (each of which can be SMP). Each cluster node has an OS running independent tasks, and nodes are connected by a communications link.

Clustering allows a node to specialize — for example, processing only certain kinds of database queries. Specialization can mean improved aggregate throughput performance when compared to the overhead of systems of multiple, unspecialized nodes. For example, a small node specializing on queries for customers "A" to "E" can often outperform a large node which has the overhead of looking at all customers. Clusters can also be used to achieve higher availability levels — a key buying criteria for the financial and information services market.

Memory-Coupling

Applications and OS's on clustered systems usually communicate using messagepassing protocols. There is overhead on sending and receiving nodes associated with preparing, sending, and waiting for a reply associated inherent in clustered systems. Further, synchronizing events on clustered systems can itself become a bottleneck — witness Digital's distributed lock manager.

Encore's reflective memory makes communications between clustered nodes behave as if tasks were communicating through memory on the same machine. Reflective memory can be shared by an entire cluster, without the contention and overhead of message-passing.

By copying application state information to reflective memory to another node, memory-coupling can also make programs more survivable and can improve overall application availability.

Single System View

System management in clusters is typically complicated by having multiple, independently operating computers. Encore's Adaptive Computing model transparently brings a single-system view of all the resources on the cluster to both programs and systems operations. The benefit is less complex (hence more reliable and quicker to implement) applications, and less operator training and intervention.

High Availability

Adaptive Computing contains several attributes which can improve availability for mission-critical applications:

- Clustering splits work over multiple nodes. If one node fails, others continue, and can restart the failed node's tasks on the survivors;
- Memory-coupling allows application restart/recovery data to be easily passed to a backup node;
- Processing nodes and I/O nodes in Adaptive Computing clusters can communicate via redundant RMS systems, and I/O can be directed through dualported disk controllers to mirrored disks. Adaptive Computing supports configurations with no single point of failure.

Market Overview

Introduction

This section examines the market for Adaptive Computing in the financial and information services industries, the two target markets Encore asked Aberdeen Group to examine. We discuss the segments of each industry, the major applications of each segment, and the opportunity for Adaptive Computing. The second portion of this section is devoted to Aberdeen's projections for market size and growth rates in the price bands for the Series 90.

Approach

Aberdeen Group concludes that a traditional, fundamental market analysis of the financial and information services industries based on historic customer buying patterns is flawed and would lead to faulty conclusions. Based on history, these markets are mature, dominated by IBM mainframes, and wedded to industry-specific applications.

In our opinion, a forward-looking approach to market analysis is needed due to a number of changed and changing circumstances:

- Post-1987 financial markets are suffering high costs and over-staffing, and many participants have been losing money;
- Financial market volumes have been peaking at higher volumes, causing the industry to add capacity to ensure the ability to handle high potential crash-day volumes;
- PCs are now economic alternatives for terminals. Future financial and information service applications will rely heavily on using the processing power of PCs, while traditional dumb-terminal applications wane;

- The volume of information available online today is staggering. Harnessing that information to gain a competitive advantage requires growing computer resources;
- Both the financial and information services markets deal in time-critical information. If the information is delayed or lost, the data is worthless — and can often cost the enterprise considerably. The ability to gather, analyze, disseminate and react (electronically) to this time-sensitive data faster than the competition directly effects market share and profits;
- Communications network bandwidth and products exist to economically allow virtually any centralized or distributed application topology;
- Line-of-business managers are increasingly demanding and winning control over their business' computing resources, often at the expense of MIS. This has led to the common-place use of computers dedicated to line-of-business applications and away from mainframes that run all of the enterprise's applications.

Financial Industries

The financial industry is composed of banking, brokerage, diversified financial services and insurance industries. They are all mature computer users in an industry where market pressures are causing worldwide consolidation and cost-cutting. Nevertheless, financial services are an attractive industry for Encore's Adaptive Computing because automation can lead to immense competitive advantages.

The financial markets are largely the preserve of outmoded mainframes and will be difficult to penetrate in general due to the entrenched competition from IBM and the other mainframers. Aberdeen Group recommends a rifle-shot approach by selling through the Advanced Technology Planning organizations present in all large financial firms rather than a general marketing campaign to the overall financial industry — there is no overall financial market, just many niches which each believes their requirements are unique.

Finance Market Key Buying Reasons

Aberdeen Group research indicates there are seven key buying reasons which are common to the financial markets, and must be carefully addressed through Adaptive Computing features and benefits. In priority order, the seven are:

- 1. Reliability - the need for the computing system to accurately and reliably complete a business transaction is paramount. The financial and business value, coupled with regulatory and fiduciary responsibilities, of financial industry transactions makes "fast and loose" computing totally unacceptable.
- 2. Availability - around-the-clock operations are common in multinational financial markets as trading and commerce goes on somewhere twenty-fours a day. System availability to service transactions is extremely important. There are numerous instances in the financial markets where the computer system is unavailable, the transaction fails and potentially huge financial obligations are incurred — for example, making bank funds transfers before the Federal Reserve closes for a long weekend. The ability to configure redundant Adaptive Computing configurations with no single point of failure is significant.
- 3. Expandability - Our research shows new applications grow at a 30% rate in the year after installation. More importantly, the fast-adapting financial industry is continually inventing and refining new services and transactions. The extraordinary expansion capabilities of Adaptive Computing are valuable to this market.
- 4. Performance - the financial industry generates high transaction volumes, among the highest of all industries. The transactions are sometimes very complex. Brokers often require the fastest possible throughput. The power and real-time speed potential of Adaptive Computing is attractive.
- 5. Investment Protection - The rush to open systems is largely away from proprietary, inflexible architectures. Adaptive Computing has numerous features which protect customer investments — even if the customer's business changes. The C language and Unix are the software protection. Merchant microprocessors and the

VME bus are the hardware protection. Lastly, the excellent expansion capacity of Adaptive Computing allows a system to grow by keeping installed hardware in place and without costly model upgrade roll-in/roll-outs.

- 6. Applications - By and large, the complexity of financial industry application software has been a high barrier to entry. Encore will undoubtedly hear at least one potential buyer say, "I wish it could run my CICS applications." However, Encore's position is not unique. There are precious few Unix-based enterprise quality applications in existence. Further, the dynamics of downsizing to open systems suggests new applications that rethink business solutions largely divorces past applications and applications approaches. Therefore, Encore must focus on delivering a strong suite of applications development tools for online, database/communications-oriented commercial applications.
- 7. Price - Pleasantly, price is low on the financial buyers list. We believe Encore can successfully sell Adaptive Computing at high margins because it is selling good — in fact, unique — value. However, the financial industry generally requires extensive service offerings as part of the price they pay.

The following sections outline the subsegments of the financial computing market. A discussion of applications is contained in Appendix A and potential application sources is contained in Appendix B.

Banking Industry

Banking is a worldwide market with concentrations in major financial centers. The two major bank segments are retail and wholesale banking. Adaptive Computing will be attractive to banks because of its ability to handle the largest share of present bank computing: online transaction processing (OLTP) and decision support.

Bank OLTP purchases began rebounding in the second half of 1989 from a slight decline in 1988 induced by low-to-no profits, and that momentum is continued into

1990. Banks are among the leaders in the downsizing trend, in spite of the predominance of mainframes.

Domestic banks are individually struggling to be survivors in a massive shakeout which will see half their current number gone within five years as the irreversible interstate banking movement rolls on. On the retail side, use of debit cards, longer hours, instant interest-rate responses to market conditions, and even more ATMs all impose more demand on existing OLTP systems for those that have them, or requires new ones for those that don't.

In wholesale banking, any bank expecting to gain or even maintain marketshare among large industrial and commercial accounts must deploy sophisticated, networked OLTP systems to compete for international lending, cash management, correspondence banking, trust accounting, mortgage lending, corporate trust and depositories.

Banks are no longer able to viably maintain their batch computer systems installed in the 1970s. Leading banks have re-evaluated their information networks and are implementing more cost-effective solutions that combine lower overall labor costs, greater security, faster response to customer requests, on-line tellers at all branches, and integration of all information on any account at one authorized point.

Lastly, banks are increasingly using automated decision support systems — often tied to a relational database — in order to make decisions more economically, efficiently, and consistent with increasing regulatory reporting requirements.

The result of these trends is that banks are now continually upgrading -- funds permitting -- their computing systems from centralized batch processors housed in dedicated data centers to on-line information networks -- representing a prime area of demand for Adaptive Computing.

Brokerage Industry

Information is the product of the securities industry. Information must be collected, analyzed, dispersed and acted upon 24-hours-per-day around the world. The global market from London-to-New York-to-Tokyo-to Zurich never rests. To catch the paperthin margins on increasingly large and risky transactions, computers trade with computers with little or no human intervention. And with such highly leveraged stakes involved, brokerage and securities firms feverishly pursue the latest in OLTP technology.

The financial community is still in shock from the market crash of October 1987. Even though the crash was partly caused by automated systems, Wall Street has renounced its renunciation of computer-based trading schemes by making them even more sophisticated. Also, the market managers added substantial capacity as a result of the unprecedented crash transaction volumes in order to forestall any market perturbation caused by lack of transaction-filling capacity.

With the industry in a protracted shakeout, it collectively sees even more automation as a way to gain individual competitive advantage. We see the following benefits of Adaptive Computing as most attractive to the brokerage industry:

- The brokerage industry has the highest degree of real-time requirements of the non-communications commercial markets. Encore has great expertise in custom realtime systems. Adaptive Computing promises excellent commercial real-time computing capabilities versus the competition;
- Most systems are proprietary and mission-critical. Encore has the reputation for custom systems and the capability to configure an Adaptive Computing system to virtually any requirements;
- Further, buyers are looking for a competitive advantage, not another off-the-shelf solution;
- Availability ranges from high to fault-tolerance. The ability to configure Adaptive

Computing systems in a redundant fashion is important, and the capability to use Reflective Memory for application checkpointing gives Encore an attractive offering for a non-fault tolerant supplier. Where fault tolerance is mandatory, it appears conceivable to be able to install FORMS on the VMS bus of a Sequoia system.

Diversified Financial Services

Diversified financial services includes those financially driven enterprises not categorized as banks or brokers. The market includes leasing, financing, and nonbank credit operations such as American Express. The diversified financial services market is attractive to Encore because it includes some of the world's largest and most respected organizations (General Electric Credit, Diner's Club, Comdisco), but also because the market generates huge volumes of high-value transactions.

Like the rest of the financial sector, the diversified financial market is increasingly cost- and efficiency-conscious, making an Adaptive Computing solution from Encore attractive.

Insurance Industry

The insurance segment is a natural for Adaptive Computing, but internal and external forces are conspiring to make it the slowest growing financial segment over the next four years (nine percent annually).

First, insurance is rapidly losing its privileged regulatory and antitrust status at a time when underwriting profits have been on a roller coaster, making bold new initiatives difficult to finance.

Second, product mixes are dramatically being realigned in face of commercial, industrial and consumer dissatisfaction. A bunker mentality is not conducive to

innovative solutions, no matter how badly they are needed to survive a just-beginning worldwide shakeout.

Third, as one of the first industries to automate, the insurance segment is nearly saturated; most recent demand was for upgrade and replacement.

On the positive side, to lower labor costs and to provide competitive bids faster, insurance companies are automating incrementally in several areas:

- They are experimenting with imaging-based technology to convert all paper data into electronic form to improve their internal efficiency;
- They are leading practitioners of cooperative processing between agencies and home office;
- Their enormous online mainframe systems must be completely rearchitected to take advantage of PCs in the client-server model. The applications are old and need replacement;
- As costs to consumers for property and medical insurance rise, regulatory pressure is enormous to reduce insurance costs, driving a squeeze on profits. Improved automation is viewed as a means of regaining traditional margins through increased efficiency.

Information Services Market

Information-for-a-fee services have matured rapidly in the 1980s, and competition is driving the search for new or improved information delivery vehicles. In order to speed full-text retrieval services, Dow Jones installed a massively-parallel Thinking Machine. Large service providers like ADP, Dow Jones, Mead Data, OAG, MCI Mail and CompuServe require up to 150 TPS from 10,000 on-line subscribers.

The kinds of computing typically done by information services — timesharing, database retrieval/display, and (increasingly) graphics/imaging — are all strong product features of Adaptive Computing. There is an emphasis on connectivity, but breadth of communications protocols is also important.

The information utility market is extremely fragmented, making general marketing difficult and potentially unrewarding. The information sold ranges from specialized fresh produce availability (a tiny niche) to generalized utilities such as CompuServe. Low cost modems and the general availability of cheap packet-switching networks has finally opened the information business to the masses.

The market is very cost-sensitive and amenable to fixed cost-lowering options such as Adaptive Computing in order to keep unit costs down. Dow Jones News Retrieval, for example, is losing customers due to recent price increases, and Nynex is shutting down its information service gateway after losing ten million dollars over the past two years. We believe cost and competitive pressures will continue.

Large information service providers frequently have multiple, networked computer systems. Our research suggests the computers are purchased based on a best-in-class buying decision at a given time. With an innovative and extensible system, Adaptive Computing has the opportunity to be the best-in-class platform for early-1990s information applications.

Perhaps the fastest growing segment of information services is the intelligent network applications of the telecommunications market, a market Encore is analyzing outside the scope of this report.

Information Service Buying Reasons

There are many similarities between the information service and financial service industries in terms of the factors used by buyers in evaluating new computer systems and vendors. Our research ranks the six key buying factors as:

- Availability
- Investment protection
- Throughput and response time
- Expansion options
- Reliability
- Price and price/performance.

Finding the Common Threads

There are common threads of customer buying needs and attitudes which must be exploited by Encore in order to maximize its marketing and product investments:

- Most financial industry and information service applications are time-critical: if the computer system is unavailable, revenue generation ceases;
- The two markets are constantly looking for innovative means to deliver lower cost services;
- Availability is a high-priority feature — and in many cases, necessity. Availability also is crucial to the telecommunications market Encore is targeting;
- Communications- and database-intensive processing are common threads. Encore must complete and then extend its communications products development program and optimize database performance for the Series 90 to take advantage of the clustering and commercial parallelism inherent in Adaptive Computing;
- Like politics, these computer markets are "local". Encore should target particular market segments, learn the customer's applications, and "be seen";
- Customer references are a critical selling tool, and should be one of the early objectives of Encore's marketing strategy;

- Both financial and information service markets rely heavily on mainframe datacenters which are relatively easy to locate and to identify the key personnel;
- Many of the large market participants are long-experienced in dealing with multiple computer suppliers. Often, best-in-class systems are purchased for specific line-of-business applications. Encore will not be excluded from evaluation per se;
- These markets today are developing new or completely rearchitecting old applications in order to gain competitive advantage — often to lower costs. Compared to mainframe alternatives, Adaptive Computing offers multiples lower costs. Compared to midrange open systems alternatives, Adaptive Computing shows excellent scaling and innovative availability features.

Market Size and Growth Rate

Aberdeen Group sizes the financial and information services markets at \$11.3 billion in 1990 for hardware and software products in the \$25K-and-above midrange and mainframe price bands which Encore has targeted for Adaptive Computing. While under-\$25K solutions (e.g. PCs) are growing rapidly, the financial and information services markets typically demand compatible solutions which scale from \$50K to several million dollars.

We project the financial and information services to grow to approximately \$20 billion in 1994. This is a highly attractive 19% annual growth rate.

Aberdeen categorizes the type of computing used in the financial markets as largely online transaction processing and decision support, while information services mixes timesharing and OLTP. While there is no inherent reason to ignore batch computing, this mode is relegated to mainframes and offer little attractiveness.

Our projections for market size and growth are depicted in the graph below.

Market Positioning

Aberdeen believes that Encore has two fundamental choices to make in positioning the Series 91 & 93 to the commercial marketplace in general. These are:

The Series 91 & 93 can be positioned as the leading price/performance multiuser commercial Risc/Unix system where it will compete directly against IBM's RS/6000, Data General's Aviion, Hewlett-Packard's Series 800, Digital's DECsystem 5000, MIPS Computing System R6000, and Sun Microsystem's SPARCserver

or

The Series 91 & 93 can be position as the next-generation products to provide enterprises with the advantages of loosely clustered systems through RMS where it will compete in the same product space (as a superior alternative computing platform, but not head-to-head) with IBM S/390 with DB2, Teradata, multiple/fiber-optic interconnected Sequent S81s, VAX 6000/9000-based VAXClusters, and Tandem.

The first alternative puts Encore back into the position it has been in with previous generations of systems — flogging price/performance to the financial and information services marketplaces that rank price/performance as relatively unimportant compared to other attributes such as availability, upgrade path, and response time.

The second positioning allows Encore to do what it does best — provide a technical sell to technical buyers. Encore can position the Series 91 & 93 as a superior alternative to traditional mainframe and midrange computing systems and charge a premium price for implementing a technically advanced architecture that will provide financial and information services enterprises with a competitive advantage.

Defying today's conventional wisdom, Aberdeen recommends that Encore initially merchandize the Series 91 & 93 for the financial and information services marketplaces based on its technological advantages to provide a higher level of

customer support than can be provided with traditional mainframes and midrange systems. The RMS-based, loosely coupled architecture that Encore internally refers to as "dirty computing" can be positioned as the next-generation architecture for fast retrieval of data from large volumes of data. This specific message will appeal to technology-driven decision makers within both financial and information services enterprises.

Other factors for recommending that Encore concentrate on an initial technology superiority theme include:

- Encore needs to capture only a very small share of the financial and information services hardware expenditures initially to be successful in terms of Encore's current nonexistent position in these marketplaces — Encore does not need to be all things to all buyers at this time;
- The advanced technology theme will reinforce Encore's commitment to its installed base of realtime customers that it will not dilute resources to focus on the general commercial marketplace;
- Encore can build on the advanced technology theme over the next three years with ongoing performance and customer success announcements. The advanced technology theme has a long life in the financial and information services industries that respect stability and continuity on the part of their mission-critical hardware suppliers;
- Encore has neither the depth of staff nor the breadth of third-party alliances to be a general systems supplier to either the financial or information services marketplaces. Therefore, traditional market positioning statements that appeal to the security blanket that many suppliers seek to promise prospects could not possibly be used by Encore;

- Encore's field sales force can quickly determine which suspects are real prospects because they have a need for the benefits of "dirty computing" and not be trapped in endless rounds of commodity price cutting against competitors that can do the same timesharing computing functions as standalone Series 91 & 93s can.

Coordination of Merchandising and Field

This merchandising plan has been created specifically with the objective of supporting the Encore field organization. The merchandising effort is intended to motivate financial and information services decision makers to consider Encore and the Series 91 & 93 as the potential basis for new information processing applications. But just having an advanced technology is not enough to capture a sale — the Encore field organization must be prepared to meet the pre-sales, post-sales and implementation support needs of financial and information services industry MIS decision makers.

Encore management should consider three field issues that need to be addressed in concert with this merchandising plan:

- The Encore field force has little experience in selling into financial and information services accounts and believes it has been misdirected several times in the past;
- Financial and information services accounts demand that their hardware and systems software supplier of record also be able to offer systems integration services to customize each specific installation. Systems integration services today is a very broad umbrella term that includes application development, networking and interoperability support for non-Encore systems, capacity planning, disaster recovery services and even strategic technology planning. In financial and information services accounts, Encore will need to lead in both coordinating the efforts of other firms and supplying support itself. Encore must be prepared to act as prime contractor;

- Encore must be careful to have completed enough field sales training and to have developed enough systems integration alliances to satisfy the customer requirements of a successful merchandizing campaign.

Proposed Series 91 & 93 Messages

Aberdeen recommends that Encore introduce the Series 91 & 93 as multiple packaged systems connected by RMS under the theme of Adaptive Computing. Encore will introduce Adaptive Computing as the style of computing for the 1990s that allows MIS managers to quickly and effectively modify their data center systems to rapidly accommodate the changing needs of their enterprises.

A sample press release might have as its theme, "First introduced to the commercial marketplace by Encore Computer in 1991, Adaptive Computing takes advantage of the best features of older technologies such as symmetric multiprocessing and Unix and transforms them into a data center system that can be easily adapted to provide users with sophisticated, new applications that require realtime response. Encore's ongoing research into reflective memory systems (RMS) has led to the technology breakthroughs required to make Adaptive Computing a reality in today's data center."

Aberdeen recommends that the minimum configuration would be a base model that we have dubbed the Encore Adaptive 291. This would consist of two dual-processor Series 91s (one for computing and one as an I/O channel), 16 MB of RMS and 4GB of DASD for \$250,000.

The specific benefits of Adaptive Computing that will be stated in Encore marketing communications are the unique abilities to:

- Search large amounts of data and update multiple applications faster than with any other system;

- Move data transparently into multiple applications at memory-speed transfer rates;
- Provide interoperability among multiple, different applications transparently through RMS;
- Add additional dedicated systems as the number of applications grow;
- Merely add processors as required to match the needs of growing applications;
- Service thousands of simultaneous online users with fast response times;
- Manage a single systems view of multiple systems;
- Distribute the system over a 3 km area;
- Create a fault-resilient system through redundant processors and RMS.

To differentiate itself from all other competitors, Encore needs to emphasize the technical capabilities and user-enterprise advantages of building data center systems around the use of RMS. Aberdeen knows of no other computer manufacturer that is following an architectural strategy that is similar to RMS. Most are caught up in merely providing the functionality on Risc/Unix systems that was previously available on CISC/Proprietary O/S ASAP and are planning the addition of Risc/Unix multiprocessing systems to be released between now and 1993.

Further, we expect a flurry of suppliers will announce mediocre cluster capabilities (including Sequent, which is using Ethernet as the cluster communications backbone). With media attention focussed on opens systems clusters, Encore has the ability to stand out with truly superior and different technology.

If Encore religiously preaches the benefits of Adaptive Computing, it has the opportunity to establish a new agenda for data center MIS executives similar to what Sybase has done with client-server for the RDBMS (relational database management systems) marketplace.

In particular, the Adaptive Computing message will reach a receptive audience within the financial and information services markets. Both marketplaces are under increasing pressures to reduce costs but, due to the time critical nature of revenue capture, be ready to expand and/or modify services instantly to meet changing user demands.

Finally, we strongly recommend that Encore not use the term "Open Systems" in its marketing messages. We believe that users now believe open systems is synonymous with "low-priced", "Unix", and "commodity box". Encore cannot make the type of gross margins it needs to grow the business if it is perceived as merely yet another open systems provider.

Product Issues

At the base level, there is very little to recommend the Series 91 & 93 computer systems to the financial and information services marketplaces. True, they are hot boxes, but Data General has already claimed the leading Motorola 88000/Unix hot box position at price levels that Encore management does not desire to be locked into. And MIS executives in the financial and information services industries rarely, if ever, buy just a hot box — they need to make long-term commitments to a supplier's current products and vision of the future.

Encore Adaptive Series Packages

Packages consisting of multiple Series 91 & 93 processors with reflective memory can be value priced to the financial and information services industries — yet allow Encore

the option of pricing basic systems components at the appropriate price levels for its traditional realtime marketplaces.

Aberdeen recommends that the base level package consist of two dual-processor Series 91s with 16 MB of reflective memory and 4 GB of DASD for approximately \$250,000. While the price can always be discounted when appropriate, this would be the minimum financial commitment that Encore should expect even for a serious trial data center system based upon a technological sale for a financial and information services mission-critical application.

In addition, Encore should place in its catalogue a range of packages up to the largest technically feasible system. For example, an Adaptive 893 consisting of 8-System 93s, 1 GB main memory, 4-64MB RMSs, and 12 GB DASD system priced at approximately \$5 million. Note that as packages become larger and more complex, the price based upon a simple parameter such as \$/MIPS should actually increase from the base level price.

Communications

The product area of greatest concern for Encore in the financial and information services industries will be communications. Both industries are heavily dependent upon communications, including IBM's SNA protocols and application specific protocols. For example, one supplier of LAN-based systems to the financial industry has over 100 engineers dedicated to providing and supporting communications protocols that are required by customers.

At a minimum, Encore must be able to support IBM's major SNA protocols, including LU2.0, LU6.2 and SNADS. In addition, we would recommend that Encore license and resell DECnet compatible networking software from Bell Atlantic's Community division if it wishes to operate in conjunction with currently installed VAX systems.

In addition, Encore should begin both developing greater in-house communications expertise and making partnerships with smaller custom consulting firms that can provide application specific protocols for customers under Encore's general supervision.

Again, Aberdeen cannot over emphasize how important the ability to supply individual customers in the financial and information services industries with the communications protocols each needs will be to Encore's success. And the issue is not necessarily the cost of the communications protocols but the ability to deliver on time, assure quality and reliability of the communications software, and support it into the future.

RDBMS

The ability to quickly develop and modify applications is extremely important to the financial industry. And the chosen vehicle to accomplish application developments is increasingly RDBMSs from the major hardware suppliers (IBM's DB2, DEC's Rdb, Tandem's Guardian and Teradata's SQL) and the independent software vendors (Ingres, Oracle and Sybase).

Encore does not have the resources to develop and support its own world-class RDBMS. Therefore, it is appropriate that Encore develop as close a working relationship as possible with all three of the major independents. Due to the multiple system and RMS architecture of Adaptive Computing, this working relationship will require expertise on Encore's part as to how specifically to develop RDBMS applications that take advantage of the unique technological capabilities of the system.

We highly recommend that Encore explore technology-sharing agreements with its two major RDBMS partners, ASK Ingres and Oracle, to optimize RDBMS performance in the clustered topology of Adaptive Computing. We believe that the I/O-managing Series 91(s) should also be executing the database read/write, logging, and archive

processes of the RDBMS. Oracle has recently announced version 6.2 with improved cluster support. The objective is to harness the distributed processing power of an Adaptive Computer with optimized DBMS performance and throughput not available from a vanilla port of the RDBMS engine. In short, the loosely-coupled RMS-based systems must be tightly-coupled from an operational standpoint and tuned for database throughput.

Lastly, we firmly believe Encore must early on demonstrate the claimed benefits of Adaptive Computing through performance benchmarks — starting with the TPC-A transaction benchmark.

Distribution Tactics

Enterprises in both the financial and information services industries that would make the commitment to Encore for what would soon be millions of dollars of systems as envisioned under the Adaptive Computing architecture would require direct sales contact by Encore for the period covered by this report.

Teradata is a similar, recent role model. Teradata's first sales were to a limited number of customers such as Citicorp and Bank of America. These sales were all direct sales. As the capability of Teradata's system became appreciated by the industry, Bull and NCR decided to OEM their systems — but not until the product had been available for approximately three years.

Even where Encore has the opportunity to form alliances with systems integrators and value-added resellers, it will be required to sell directly to financial and information services MIS executives itself to establish credibility over the planning period.

As a result, we recommend that Encore hire sales representatives who have experience in selling to the financial and information services industries. Specialized sales reps have the potential of shortening the amount of time (by at least 12 months)

that it would take Encore to make the initial key introductions with decision makers in both the financial and information services industries.

Competitive Overview

The Series 91 and 93 packaged as Adaptive Computing will have to compete against both established suppliers and other new starts in both the financial and information services marketplaces.

Aberdeen research indicates that the technological leader to beat in the financial services industry will be the IBM S/390 running MVS/ESA with applications developed under CICS for DB2. In the information systems marketplace, the major technological competition will be Tandem Computers providing 24-hour availability and extensive communications support. In addition, Digital Equipment Corporation's VAXcluster, Teradata and Sequent will represent competitive alternative technological threats to Encore's Adaptive Computing.

IBM S/390 and DB2

IBM continues to position the high-end S/390s running MVS/ESA with applications developed using CICS and targeted for its DB2 database as the most powerful system available for financial institutions' data centers. (The multi-million dollar acquisition and support costs are too high to even bother pitching to most information services enterprises.)

Aberdeen believes that IBM is developing its own version of Adaptive Computing — but instead of RMS it will employ a dedicated Risc-based multiprocessor database engine running DB2 that will service multiple S/390s.

If Encore can conclusively show that it can outperform a high-end S/390-720 for commercial computing, Aberdeen believes that sales opportunities could easily surpass Encore's most optimistic expectations.

Encore's strengths: Low hardware and support costs, expandability, excellent price/performance, choice of RDBMSs, application interoperability — all equalling the ability to give the financial institution that successfully implements Adaptive Computing a competitive advantage against those who stay with high-cost, inflexible S/390s.

Encore's weaknesses: Limited support resources, minimal communications capabilities, and lack of credibility in the financial marketplace.

Tandem

Tandem is continuing to make inroads into the information services. Information service decision makers are especially attracted to Tandem's high-availability continuous processing capability, range and quality of communications protocols supported, and modular expansion growth path.

Tandem is currently migrating from its proprietary architecture to a higher performance system based upon MIPS Computing Systems Risc processors and Unix. We believe that this product transition will not be complete during the planning period and we believe that this effort is absorbing any engineering resources that Tandem may have to replicate the functionality of Adaptive Computing within the planning period.

Tandem derives almost half its product revenues from the financial markets. It is well-established with excellent references, its sales force is well-versed in the industry's nuances, and fault-tolerance is prized for certain applications.

Encore strengths: Low hardware and support costs, excellent price/performance, extremely fast end-user response even when searching through large amounts of data. Encore's Adaptive Computing has the opportunity to be the low-cost, highperformance database search and retrieval system to installed front-end Tandem systems.

Encore's weaknesses: Limited support resources, minimal communications capabilities, no fault tolerance, and lack of credibility in the information services marketplace.

DEC VAXcluster

DEC's VAXcluster has set the technological leadership role for loosely-coupled systems for the last decade. Controlling the individual VAXes and disks in a VAXcluster is a device known as a Star Coupler on smaller clusters or an HSC controller on larger clusters. These cluster control devices are dedicated computers. Encore's technological leapfrog is the use of RMS to perform many of the same functions as a cluster controller but much faster.

DEC will claim that Encore's Adaptive Computing is just a pale attempt at imitating the functionality of a VAXcluster. Aberdeen recommends that Encore respond by claiming that Adaptive Computing is a generation of technology beyond the VAXcluster and fulfills the functional specifications that DEC claimed for VAXcluster but never could effectively deliver (i.e: single-systems view, fast access to data located anywhere within the cluster, and node growth with no end-user response degradation).

Encore strengths: Unix systems in a cluster, faster end-user response time, low hardware and support costs, excellent price/performance, beginning of new generation of systems while DEC is phasing out VAX, and interoperability of applications.

Encore weaknesses: Minimal communications capabilities.

Teradata

Teradata was started to provide Citicorp with a means to access large amounts of data for decision support applications. The Teradata system is composed of multiple Intel 80X86 processors that are acquired as modules to perform specific tasks such

as I/O and processing. In addition, the Teradata system has an integrated SQL RDBMS. Note that the typical Teradata system is configured at \$1M-\$1.2M and then has \$.5M of upgrades over the next 18 months.

Encore will find that Teradata has already made inroads into the largest banks for data retrieval applications. And many prospects within the financial services industry will initially perceive the Adaptive Computing architecture as being in the same class as the Teradata. Their first question will be whether the Series 91 & 93 can provide their enterprises with the ability to run OLTP applications that Teradata has promised but never been able to deliver.

Encore strengths: Industry-standard RDBMS with Ingres that is far superior to Teradata, already on the Risc performance curve that Teradata must jump to eventually, ability to grow by adding full systems — not just processors and modules, price/performance.

Encore weaknesses: Minimal communications capabilities.

Sequent

Aberdeen has found Sequent in the financial services institutions as a potential replacement for large mainframes. Technologists within these institutions explain that multiple Sequents could be connected via high-speed fiber optic cables to create a single-systems view of an OLTP application. In this scenario, each Sequent would handle one portion of the application and communicate with the others only when there was a need for a cross reference check with a database managed by one of the other systems. The advantages are the use of a standard RDBMS, growth path, ease of modifying applications compared to a mainframe running CICS, and much lower cost.

The large majority of Sequent's sales are for medium-sized applications where the user needs more power than a microcomputer such as an Altos or NCR Tower but does not want to pay the price of a DEC VAX. As a result, Sequent has not announced a vision of high-end computing comparable to Encore's Adaptive Computing.

Encore strengths: RMS for interoperable applications and fast response time compared to Sequent's interconnection with fiber optic cable, and RDBMS pointers and higher-performance Risc processors compared to Sequent's CISC Intel 80X86. Encore weaknesses: None.

Sales Program Recommendations

Recognizing that Encore will have a limited budget to introduce Adaptive Computing, we would make the following recommendations:

- Initial awareness among prospects can best be created through a PR campaign that is aimed at the general technical and specific financial technical journals such as Wall Street Computing Review;
- Encore should pre-package 1/2 day seminars that can be presented in-house to financial and information services technical executives on what Adaptive Computing is, how it can be implemented, and the advantages that it can bring to users. This seminar should be ready for the field upon the announcement of Adaptive Computing;
- Four months after introduction and after the Encore sales force has been trained, Encore should follow-up with a direct mail campaign to the VP's of

Technology of major financial and information services enterprises offering to provide the in-house seminar on Adaptive Computing;

- After Encore has had time to learn MIS executives' response to Adaptive Computing, Encore should prepare a 15-minute video tape that stresses the advantages that customers most respond to and answers the most commonly raised objections. This tape should be used as a leave-behind reminder as part of initial sales calls.

Appendix A

Financial Applications

Banking Applications (Retail)

On-line banking - integrates demand deposit accounting (checking), deposit instruments, installment and mortgage loans, and customer relationship data. Application software is a prerequisite in this market. In the U.S., commercial banks have not implemented on-line banking and are unlikely to in the next five years. There is no pressing market demand to do so. U.S. savings and loans have traditionally been on-line at the teller level and have implemented automated teller machines for off-premise services. There are 25 large S&L's requiring more than 50 TPS for their systems. However, the problems of S&L's are widely known.

Internationally, banks in Europe and Japan are more centralized an on-line than their U.S. commercial counterparts. Very large mainframes are the norm, requiring up to 125 TPS. In Europe, the 1992 economic changes will probably loosen the national computer vendor strangle-hold on large bank computer systems, creating an opportunity.

Relationship accounting - customer information systems aggregate data from individual bank applications into a relationship repository. The intent is to allow personnel to view all of a customer's activities at the bank. While widely desired, these systems have only been installed at the largest banks — or else the

functionality is rudimentary. The decision support characteristics of a relational database are ideal for this application, but RDBMS performance has been so poor (until recently) that these systems could not be price/performance justified. While application performance requirements in TPS are modest, the processing necessary to manage complex decision support queries in real-time is high.

Automated teller machines - local banks require up to 15 TPS. Nationwide ATM networks like CIRRUS require 25 TPS or less. This market is maturing. Volume requirements are higher, to 60 TPS, in Europe due to the widespread use of debit cards for electronic funds transfer (EFT). The darling market of the 1980s, we do not expect future growth at anything close to the historical rate.

Credit and debit card - This market demands high throughput, expandability, and very high availability. Card processing is quite lucrative because financial institutions charge a transaction fee on every one of the millions of transactions processed daily. In spite of the strength of the entrenched card processors, Mastercard and Visa, Sears started Discover in the mid-1980s and AT&T's wildly successful Universal card broke into the business in 1989. We do not believe market-entry restrictions by Mastercard and Visa will stop other new competitors.

Card applications have typically required multi-million dollar mainframes with faulttolerant front ends. Volumes range from 100 TPS regional nodes to 300 TPS central switches. Adaptive Computing has the benefits of excellent price/performance, mainframe-caliber throughput — plus, high availability, and good connectivity.

Banking Applications (Wholesale)

International funds transfer - large money-center banks generate 10K-15K transactions per day (less than 1 TPS) in this high-profile application. Security and availability are key requirements.

Cash management - allows corporate cash managers the ability to move monies around the globe. Timing is critical, volumes are a modest 20 TPS in large systems, and information retrieval and presentation is a critical factor in these applications.

Foreign exchange - purchase and sale of foreign currencies. Up to 40K transactions per day (1 TPS).

Interbank clearing systems - time-critical bank-to-bank item transfer of checks, drafts, and notes among members. Volumes to 200 TPS peak. Examples include CHIPS (the NY clearing house) and FedWire, which handles Federal Reserve Bank transactions. While this applications appears very attractive, there are only a handful of clearing house systems in place — it is not an attractive market.

Brokerage/Securities Applications

Retail trading - multiple computers manage the flow of information from the broker's desktop through wire rooms to the exchange floor and back. Volumes are substantially governed by the retail broker's ability to enter transactions; 75 TPS is a peak load at a large brokerage firm.

Mutual funds - shareholder systems and securities accounting system volumes are up to 100 TPS. The stock ticker plant application (usually on a fault-tolerant computer) maintains real-time database and position information for the institution's wholesale business. At the start-of-day, opening quotes drive volume to 150 TPS. These transactions require minimal database activity.

Program trading - about 400 firms engage in program trading. A typical program involves watching a basket of the Standard and Poor's 500 stocks and making continual mathematic calculations against the underlying options. This application requires the ability to maintain the security "basket" database, run the linear equations

necessary to make the buy/sell determination, and execute the trade through an automated interface to the trading floor. This can involve the quick purchase or sale of 500 stocks; each stock is an individual transaction. Because program traders are competing with other large institutions, speed and accuracy become competitive advantages.

Commodity trading - foreign currency, bond, and commodity trading involves international markets with intense competition. Individual institutional traders use rapidly changing spreads between markets to make minuscule profits (or losses) on large volumes of transactions. Computers are increasingly used to manage multiple market databases, determine profitable spreads, and signal traders to initiate the transaction. Volumes range to 75 TPS.

Information providers - Reuters, Telerate and other securities information providers deliver securities market information over proprietary networks to customers. The systems require numerous (over 50) information input sources and the capacity to communicate 1-2M bytes per second over up to 2,000 communications lines.

Insurance Applications

Claims Processing systems - these custom-tailored applications at large insurance companies generate volumes to 60 TPS. None have been implemented with an RDBMS to our knowledge. Aberdeen field research shows numerous insurance companies are reimplementing this core application — a once-in-a-decade opportunity.

Agent Support - agent/broker online support for customer records, policy information, price quotes, etc. Volumes to 40 TPS against many databases.

Risk Management - online ad hoc management of claims data to adjust risks to rates.
Large databases with decision support query volumes.

Policy Systems - inquiry, billing, payments and renewals with online access by local or remote agents. Volumes to 35 TPS. Highly customized or sourced from specialized ISMs such as Policy Management Systems or Continuum.

Actuarial - this computation-intensive application has traditionally been done on large mainframes in batch or background. An opportunity exists for on-line actuarial decision support.

Imaging - we believe the insurance industry will increasingly embrace digital imaging and the use of object-oriented databases. The on-line storage and retrieval capacities of these systems at the major insurance companies are conjectural but have the potential to be very large. Further, our research shows the insurance industry moving towards implementing imaging systems over the next four years. Appendix B

Industry Application Sources

Company Name &	Application Name	Description	Address
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Santa Cruz Operations Suppliers

Dynamic Decisions, Inc. based financial trading 134 W. 26th Street 4th Floor New York, NY 10001 212-242-0108	CYBORG/FACTS software	Unix and VAX/VMS	
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IMS Business System Corporation 29 West 30th Street 8th Floor	BANK STREET	Retail Banking System	
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New York, NY 10001
212-563-5255

The Peterjohn Co. The Reporter Mortgage Credit Reporting

781 Pearson Street
Des Plaines, IL 60016
312-583-8129

Sharp Data Systems ACCESS Bank Retail Banking System
Accounting System

814 S. Lamar #22
Amarillo, TX 79106
214-522-4508

Sistematica Ltda. S.F.T.C. Retail Banking System

Branch Office Automation Carrera 7a.

NR 72 - 64 Int. 30

Bogota

Columbia

9-011-571-2114400

Software Conquerors Document Clearing Retail Banking System

Tracks documents,

1416 W. 16 Street checking deposits

Davenport, IA 52804 319- and balances

323-2440

Computoservice, Inc. INTELIX, Plant Communications,

Records System, telephone

reporting and

1315 Stadium Road Toll Rating maintenance software.

Mankato, MN 56001 Services, Trouble

507-625-1691

The Brookside Corp. Brookside Money Brokerage System

Entry, calculation,

Blackstone Centre reporting of transactions

Suite 201

Six Blackstone Valley Place

Lincoln, RI 02865
401-333-9580

Computer Design Financial Financial Cash Management,
Portfolio Portfolio Valuation
10-12 Exhibition Road Management
South Kensington
London, SW7 2HF England
01 5897331

Anasazi, Inc. AlisaHotel Reservations
7500 N. Dreamy Draw
No. 170
Phoenix, AZ 85020 602-870-
3330

Data General Suppliers

Software Alliance Corp. Total Banking Retail Banking System
Solution (TBS)
2150 Shattuck Avenue
Berkeley, CA 94704
415-548-7752

American Real-Time ARTS Quote, Equities Terminal
Services Client Tell Broker Productivity System
71 Broadway
New York, NY 10006
212-742-1100

Financial InformationThe Fin System Automation of Stock
Network Portfolio Management

6656 Valjean Avenue
Van Nuys, CA 91406
818-782-0331

International Treasury Viewpoint - FX Real-time Foreign
Systems, Inc. Exchange Trading System

5 Middlesex Avenue
Somerville, MA 02145
617-776-1779

Merrin Financial Merrin Financial Online Trading System
Trading System 325 Fifth Avenue
New York, NY 10016
212-684-7333

Zaitech Software, Inc. ZAI*NET Foreign Exchange and Commodity Trading
26 Broadway, Suite 400 System
New York, NY 10004
212-837-7872

Encore Computer and AberdeenGroup Confidential Page
Company Name & Application Name Description
Address
Hewlett-Packard Applications

B/Systems Credit Application Credit Application
Development Corp. Processing System Processing Software
OS = DOS, MPE, VMS,
One Greentree Center HP-UX
#103
Marlon, NJ 08053
609-596-6777

Flextell On Line Account
Flextell Systems, Inc.
Processing System
P.O. Box 428655
Cincinnati, OH 45242
513-489-6246

Integrated Information CUA/3000 OLTP System for Management Systems
Organizations Similar
to Federal Credit Unions Rockland
Professional Center #100
1940 N. 13th Street
Reading, PA 19604
215-921-9141

Summit Information Spectrum 3000 System for Credit Unions
Systems Corporation ATM processing On
Line Collections, Loan
850 SW 35th Street Application and Tracking
Corvallis, OR 97339
800-937-7500

First Total Systems, Inc. The Total Bank Integrated Data Processing
System System for Community
Banks

1325 S. Colorado Boulevard #24
Denver, CO 80222
303-753-0295

Bea Management PARTS Foreign Currency
Software Transaction Software

150 E. 58th Street
38th Floor
New York, NY 10155
212-755-4700

Qusim Systems, Inc. PERFORMANCE- Portfolio Trading
ONESystem
1301 Shoreway Road #325
Belmont, CA 94002
415-982-4622

Linian Systems, Inc. PF+ - Portfolio On Line Mutual Fund
Management Accounting and
716 Gordon Baker Road System Reporting System
#217
North York, Ontario M2H 3B4
Canada
416-493-3133

Software Systems Security/Pension Mutual and Pension Fund
Technology, Inc. Management
Transaction Software

305 Broadway, 4th Floor
New York, NY 10007
212-964-9600

Belvedere Systems The Trading System On Line Trading System

321 Columbus Avenue
Boston, MA 02116-5713
617-353-0800
88 Open Applications

Aberdeen is unaware of any financial services or information utility applications with
the 88 Open standard based on the latest application catalog. International
Business Machines RISC System/6000

Decision Software, Inc. Decision Software Security Trading
Trading SystemSystem
116 John Street(DSTS)
Suite 308
New York, NY 10038
212-385-1662

FD Consulting, Inc. Real-Time MIPS Securites Market
Information System
22 Cortlandt Street
8th Floor
New York, NY 10007

212-766-1420

Financial Database CAESAR Securities Trading
Systems Accounting System

1155 Frazee Road
Suite 800
San Diego, CA 92108
619-491-3110

FXD Telerate, Inc. The FXD/Telerate Securities Trading
Trading SystemSystem
2091 Landings Drive
Mountain View, CA 94043
415-961-0484

Teknedron Software The Teknekron Securities and Foreign
Systems, Inc. Trading SystemCurrency Trading System

One Exchange Plaza
New York, NY 10006
212-483-0153

VI Corp DataViews (R) Data Analysis Software

Amherst Research Park
160 Old Farm Road
Anherst, MA 01002
413-253-3482

Control Data Corp. FOCUS XP Credit Union Software

Credit Union Services
700 W. Hamilton Avenue
Campbell, CA 95008
408-866-3003

Decision Systems, Inc. Info-Lease Lease Portfolioa
Management System
75 S. Fifth Street
Mezzanine
Minneapolis, MN 55402
612-338-2585

DISC, Inc. DISC ACCESS Cash Management System

1314 Bedford Avenue
Baltimore, MD 21208
301-486-0410

Dyatron Corporation Bankruptcy Monitoring Bankruptcy Management
System System
Mortgage Systems Div.
8669 Baypine Road Dyatron Mortgage Loan Servicing
Suite 101 Servicing System Automation System
Jacksonville, FL 32256
904-737-4177 Foreclosure Claims Foreclosure Claims
System Preparation System

Foreclosure Monitoring Foreclosure Activity
System Tracking System
Financial CLASS Loan Servicing System
Automation Ltd.
CLEO Consumer Loan
230 W. Monroe Street Evaluation System
Suite 800
Chicago, IL 60606 RLOTS Loan Origination & 312-641-
2070 Tracking System

First National FIN/ESS Online Financial &
Systems Corp. Accounting System for Banks,
S&Ls, Credit Unions.
Ten Main Street
P.O. Box 1390
Cotuit, MA 02635
508-428-3600

IFS International Technique Plus II Front-end, Node Processing
System for Financial
3033 S. Kettering Boulevard Institutions, Retailer, etc.
Suite 311
Dayton, OH 45439
513-296-1116

IMS Business BANK STREET I International Banking System
System Corporation

29 West 30th Street
New York, NY 10001
212-563-5255

Integrated Analytics MarketMind Securites Market
Corporation Analysis System

13315 Washington Boulevard
Suite 301
Los Angeles, CA 90066
213-578-5052

Investment Support FUTRAK (RS/6000) Securities Trading Risk
Systems, Inc. Management System

1455 Broad Street
Bloomfield, NJ 07003
201-338-0321
LeaseTeam, Inc. LEASE PLUS Portfolio Management and
Accounting Software
407 S. 27th Avenue
Omaha, NE 68131
402-346-4600

National Investor NIDS/Fund Mutual Fund Accounting & Data
Services, Inc. Portfolio Management System

33 Flying Point Road NIDS/Port Portfolio Management System
Southampton, NY 11968
516-283-1100 NIDS/Share Transfer Agency and

Shareholder Servicing System

Synthetic Mortgage Series Mortgage Brokerage and
Intelligence, Inc. Servicing System

286 Fifth Avenue
New York, NY 10001
212-967-2399

Health Systems Managed Care Feedback Health Insurance Claim
Design Corporation Monitoring System

1330 Broadway
Suite 1135
Oakland, CA 94612
415-763-2629

Insurnet, Inc. The Insurnet System Insurance Agency Automation
System
1900 Powell Street
Emeryville, CA 94608
800-227-0796

Travelers/EBS, Inc. Independence Insurance Agency Management
System
2701 Maitland Center Parkway
Maitland, FL 32751
407-660-2550

Encore Computer and Aberdeen Group Confidential Page Appendix C

Accompanying Oral Presentation

2024 Metadata

Summary of Findings

The document is a comprehensive market strategy report prepared by Aberdeen Group for Encore Computer Corporation, focusing on the Series 91 and 93 systems. The report outlines the market opportunities, recommended strategies, key positioning messages, product issues, distribution program recommendations, competitive overview, and sales program recommendations for these systems in the financial and information services industries. The report emphasizes the concept of "Adaptive Computing," which leverages technologies like symmetric multiprocessing, clustering, memory-coupling, and high availability to offer superior performance and flexibility compared to traditional midrange and mainframe systems.

Recommendations for Computer Technology Buyers

The document advises computer technology buyers, particularly in the financial and information services sectors, to consider Encore's Series 91 and 93 systems due to their advanced technological capabilities. Buyers should focus on the benefits of Adaptive Computing, such as high availability, superior expansion, investment protection, and better price-performance ratios. The report suggests that buyers look beyond commodity RISC/Open Systems and evaluate the unique advantages offered by Encore's Reflective Memory Systems (RMS) and Adaptive Computing architecture.

Analysis of Technology

The technology discussed in the document, particularly Adaptive Computing, is highly relevant to mainstream computing in 2024. The concepts of symmetric multiprocessing, clustering, and memory-coupling are foundational to modern cloud computing and distributed systems. High availability and fault tolerance, emphasized in the report, are critical features in today's data centers and enterprise IT environments. The document's focus on flexible, scalable, and high-performance computing solutions aligns well with current trends in edge computing and hybrid cloud architectures.

Historical Impact on Companies

The contents of the document played a significant role in shaping the strategic direction of Encore Computer Corporation. By focusing on Adaptive Computing and targeting the financial and information services markets, Encore aimed to differentiate itself from competitors and establish a new architectural standard. Although the company faced financial challenges and eventually ceased operations, its innovations in parallel processing and memory-coupling influenced the development of high-

performance computing systems. The report also highlights the competitive landscape of the early 1990s, providing insights into the strategies of companies like IBM, Tandem, and Sequent.

Market Acceleration Catalysts

1. Adoption of Adaptive Computing in financial and information services industries.
2. Emphasis on high availability and fault tolerance in mission-critical applications.
3. Integration of Reflective Memory Systems for superior data transfer and application interoperability.
4. Development of scalable and flexible computing solutions to meet evolving enterprise needs.
5. Strategic partnerships and alliances to enhance system integration and support services.
6. Focused marketing and sales efforts targeting technology-driven decision-makers.
7. Performance benchmarks and demonstrations to validate the benefits of Adaptive Computing.
8. Expansion of direct sales force with industry-specific expertise.
9. Continuous innovation in clustering and multiprocessing technologies.
10. Leveraging advanced RDBMS capabilities for high-performance data management.

Keywords

Adaptive Computing, Reflective Memory Systems, symmetric multiprocessing, clustering, high availability, financial services, information services, Encore Computer, market strategy, parallel processing.

Citations:

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